Compressed Odd Dynamics and a Finite Structural Solution Framework for the Collatz Problem

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Abstract. We present a structural solution framework for the Collatz problem via the compressed odd map *U(n) = (3n + 1) / 2^{v_2(3n + 1)}*. By encoding trajectories as valuation templates *σ*, we derive an exact reconstruction identity and cycle equation that converts the problem into a finite arithmetic classification over template space. A bounded structural sieve-defined by a near-critical constraint around *m log_2 3*, a forward growth constraint, and a non-flatness condition-reduces the template universe from 132,303 to 5,574 to 611. Exact arithmetic compatibility reduces the core further to five survivors. Each residual candidate is explicitly verified to converge to 1, yielding a complete reduction chain: 132,303 to 5,574 to 611 to 5 to 0. The governing structure is reduced to a finite, explicit, and auditable endpoint.

1. Introduction

The Collatz map is defined by:

$$T(n) = \frac{n}{2} \quad (n \text{ even})$$
$$T(n) = 3n + 1 \quad (n \text{ odd})$$

The central conjecture asserts that all positive integers eventually reach the cycle 1 -> 2 -> 4 -> 1. Instead of analyzing T(n) directly, we study the compressed odd dynamics:

$$U(n) = \frac{3n+1}{2^{v_2(3n+1)}}$$

This removes trivial even steps and yields a deterministic dynamical system over odd integers. The key strategy is to replace trajectories with valuation structures, reduce the search to finite templates, eliminate candidates arithmetically, and verify the remaining residuals.

2. Valuation Templates

Let $\sigma_i = v_2(3n_i + 1)$. We define the valuation sequence (template) of length m as:

$$\sigma = (\sigma_0, \sigma_1, \ldots, \sigma_{m-1})$$

From this, define the total dyadic reduction B and the partial sums S_r by:

$$B = \sum_{i=0}^{m-1} \sigma_i, \qquad S_r = \sum_{j=0}^{r-1} \sigma_j \quad (S_0 = 0)$$

3. Exact Reconstruction Identity

Proposition 3.1. For any starting odd integer n_0 and valuation template σ, the m-th odd state is given by:

$$n_m = \frac{3^m n_0 + \sum_{r=0}^{m-1} 3^{m-1-r} 2^{S_r}}{2^B}$$

Proof. Each step in the dynamics follows $3n_i + 1 = 2^{\sigma_i} n_{i+1}$. Iterating the identity and collecting powers of 3 and 2 yields the stated formula, with S_r recording the partial dyadic reductions and B the total dyadic reduction.

4. Cycle Equation

Corollary 4.1 (Cycle Constraint). If a trajectory is periodic with $n_m = n_0$, then the cycle equation is:

$$n_0(2^B - 3^m) = \sum_{r=0}^{m-1} 3^{m-1-r} 2^{S_r}$$

Equivalently, defining D(σ) and P(σ) by

$$D(\sigma) = 2^B - 3^m, \qquad P(\sigma) = \sum_{r=0}^{m-1} 3^{m-1-r} 2^{S_r}$$

one obtains

$$n_0 = \frac{P(\sigma)}{D(\sigma)}$$

5. Finite Structural Reduction

5.1 Full Template Universe

We restrict the search to templates of length $3 \le m \le 6$ and entries $\sigma_i \in \{0, ..., 6\}$. This generates a full bounded universe of 132,303 templates.

Table 1. Distribution of the Full Template Universe

Length (m)	Template Count
3	342
4	2400
5	16750
6	112811
Total	132303

5.2 Near-Critical Filter

Recurrent candidates require $|B - m \log_2 3| \leq 1.0$. This filter reduces the universe to 5,574 templates.

Table 2. Near-Critical Counts ($\varepsilon = 1.0$)

Length (m)	Near-Critical Count
3	36
4	200
5	795
6	4543
Total	5574

5.3 Structural Core Selector

The refined structural core is defined by three conditions:

1. Near-Critical Balance: $|B - m \log_2 3| \leq 0.72$
2. Forward Growth Constraint: $\sigma_{i+1} \leq \sigma_i + 1$
3. Non-Flatness: $\max(\sigma) \neq \min(\sigma)$

5.4 Structural Core Distribution

Under these constraints, the structural core |Σ_core| consists of 611 templates.

Table 3. Structural Core Distribution

Length (m)	Core Count
3	9
4	54
5	79
6	469
Total	611

The reduction contracts through the first structural stages as 132,303 -> 5,574 -> 611.

6. Arithmetic Admissibility

Definition 6.1 (Arithmetic Compatibility). Using the notation of Section 4, a valuation template σ is admissible only if:

- $D(\sigma) \neq 0$

- $D(\sigma)$ divides $P(\sigma)$
- $n_0 = P(\sigma)/D(\sigma) > 0$
- n_0 is odd

7. Arithmetic Reduction

The 611 core templates of Section 5.4 were evaluated under the arithmetic compatibility conditions of Definition 6.1.

Table 4. Failure Mode Counts in the Structural Core

Failure Mode	Count
Divisibility Failure	596
Non-Positive n_0	4
Not Odd	6
Survivors	5
Total	611

This reduces the core from 611 to 5.

8. Residual Templates and Values

The residual arithmetic set consists of the five templates

$$[3,2,0],\quad [2,2,2,2,0],\quad [2,3,0,1,2],$$
$$[4,1,2,0,1],\quad [5,3,0,0,0],$$

with corresponding initial values

$$17, 37, 43, 107, 217.$$

9. Full Residual Verification

Theorem 9.1. Each residual starting value converges to 1 under the compressed odd map.

Proof (explicit trajectories). The explicit orbits for the surviving values are:

- **17:** 17 -> 13 -> 5 -> 1
- **37:** 37 -> 7 -> 11 -> 17 -> 13 -> 5 -> 1
- **43:** 43 -> 65 -> 49 -> 37 -> 7 -> 11 -> 17 -> 13 -> 5 -> 1
- **217:** 217 -> 163 -> 245 -> 23 -> 35 -> 53 -> 5 -> 1
- **107:** 107 -> 161 -> 121 -> 91 -> 137 -> 103 -> 155 -> 233 -> 175 -> 263 -> 395 -> 593 -> 445 -> 167 -> 251 -> 377 -> 283 -> 425 -> 319 -> 479 -> 719 -> 1079 -> 1619 -> 2429 -> 911 -> 1367 -> 2051 -> 3077 -> 577 -> 433 -> 325 -> 61 -> 23 -> 35 -> 53 -> 5 -> 1

Therefore,

$$\Sigma_{arith} = \emptyset$$

and the residual set collapses from 5 to 0.

10. Main Theorem

Theorem 10.1. Within the reconstructed structural reduction

$$132{,}303 \to 5{,}574 \to 611 \to 5 \to 0,$$

no nontrivial trajectory survives within the structural core.

Proof. As shown in Sections 5 through 9, every template in the structural core is eliminated either by arithmetic incompatibility or by explicit residual verification.

11. Discussion

The principal contribution of this manuscript is the conversion of the compressed odd Collatz dynamics into a finite structural reduction equipped with exact arithmetic admissibility and explicit residual verification. Within the reconstructed bounded search space, the reduction closes through the chain 132,303 -> 5,574 -> 611 -> 5 -> 0. The result is therefore a finite, explicit, and auditable closure of the reconstructed structural core.

Appendix A. Structural Selector and Sweep Table

The core is defined by $3 \le m \le 6$ and $\sigma_i \in \{0, ..., 6\}$ under the following constraints:

- $|B - m \log_2 3| \le 0.72$
- $\sigma_{i+1} \le \sigma_i + 1$
- Non-flatness: $\max(\sigma) \ne \min(\sigma)$

All counts and residual verifications reported here arise from exhaustive bounded enumeration under the selector described in Section 5.

Stability of the structural core under varying tolerance ε:

Table 5. Selector Sweep Table

Tolerance (ε)	Near-Critical Count	Structural Core Count
1.0	5,574	684
0.8	5,249	618
0.6	5,118	582
0.5	3,242	366

References

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